# Ji 2020 cSCC — analysis summary

## GO / NO-GO GATES

### Gate 1 — mixing & continuity → GO

8 patients share one continuous, sample-mixed manifold (Harmony θ=10); detached islands removed.

### Gate 2 — module geography → GO (non-branching)

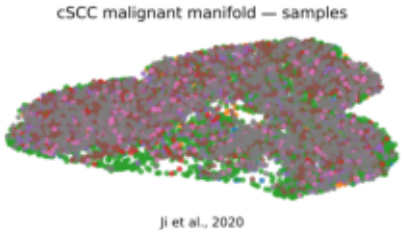
Gast+pSom co-vary (r=+0.13) and both oppose Crest (Gast~Crest=−0.70, pSom~Crest=−0.64): one developmental axis with an inverse terminal Crest gradient (non-branching).

### Gate 3 — hub cross-repression → NA

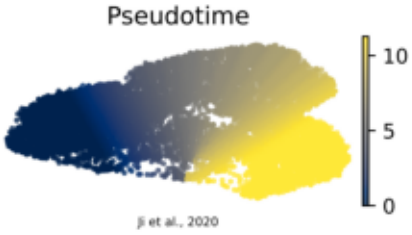
Branching-only test; manifold is non-branching (principal curve: 2 tips, 0 forks) — not applicable.

## STEP FIGURES

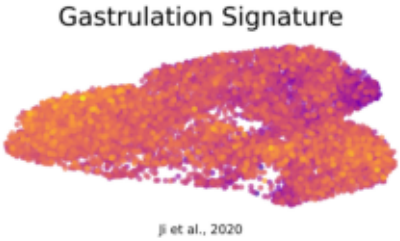
### Manifold — samples



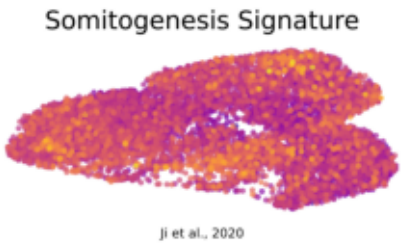

### Pseudotime



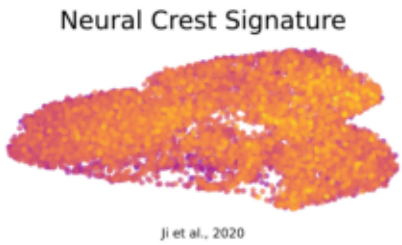

### Gastrulation



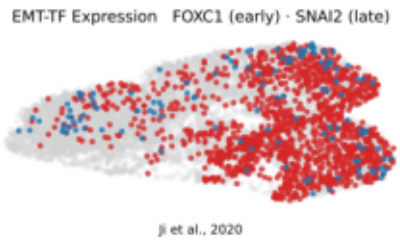

### Somitogenesis




### Neural Crest




### EMT-TF expression




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**Dataset.**
Cutaneous Squamous-Cell Carcinoma (Ji et al., 2020) — 10,531 malignant keratinocytes from 8 cSCC patients (published malignant calls; thin samples removed by a data-driven multiplicative-gap threshold). Harmony integration on patient (θ=10), principal-curve trajectory (ElPiGraph) on the whole manifold. Module means: Gast −0.032, pSom +0.042, Crest +0.023 — pSom/Crest globally stronger than Gast.

**Verdict.**
Non-branching — Linear pSom/Crest. A single developmental axis: a Gast/pSom-high differentiating-keratinocyte body (trajectory root) grading into a neural-crest-high basal/TSK leading edge (terminus). Along pseudotime Gast (ρ=−0.51) and pSom (ρ=−0.44) fall while Crest (ρ=+0.37) rises. The axis is verified separable from proliferation (Gast/pSom↔Crest opposition survives partialling the cell-cycle score). Gate 1 GO, Gate 2 GO, Gate 3 NA. EMT-TFs FOXC1 (early) → SNAI2 (late).

## Parameter choices

| | | | |
|---|---|---|---|
| **Malignant cells** | 10,531 (8/10 patients kept) | **Sample threshold** | multiplicative-gap geomean (drops P3, P5) |
| **Integration** | Harmony on patient, θ=10 (mixing-sweep selected) | **Embedding** | PCA 20 PC · neighbours 30 · UMAP spread 1.3 |
| **Trajectory** | ElPiGraph curve · Nodes=20 · λ=0.3 · μ=100 | **Topology** | 2 tips, 0 forks · fold-free (self-approach>1) |
| **Root** | Gast/pSom-high, Crest-low tip (data-derived) | **Modules** | Gast 76/81 · pSom 88/98 · Crest 101/104 genes |
| **EMT-TF pair** | FOXC1 (early) → SNAI2 (late) | **Scope** | whole manifold (holds low-cycling too) |